Urban Cycling Adoption and Infrastructure Investment: Evidence from Twelve European Cities, 2014-2024

Hartwell, J. and Okafor, A. — Cambridge Centre for Transport Studies, 2025

Abstract

This study examines the relationship between cycling infrastructure investment and modal share growth in twelve European cities over the period 2014 to 2024. Drawing on municipal transport data, household travel surveys, and capital expenditure records, we find that cities investing more than 8 percent of their transport capital budget in cycling infrastructure exhibited modal share growth of 4.2 percentage points on average, compared with 0.8 points for cities investing less than 3 percent. The relationship is non-linear: cities crossing a critical threshold of approximately 50 kilometres of protected cycle lane per 100,000 residents experienced disproportionate adoption gains, suggesting network effects matter more than incremental additions. The findings have implications for transport policy, particularly in cities where cycling currently represents under 5 percent of trips.

1. Introduction

Cycling has re-emerged as a serious component of urban transport policy across Europe over the past decade. Concerns about urban air quality, climate obligations, public health, and congestion have prompted municipal governments to commit substantial resources to cycle infrastructure. Yet the relationship between investment and behavioural change remains poorly understood, and policymakers face genuine uncertainty about how to allocate scarce capital across competing transport priorities.

This study analyses outcomes in twelve cities selected to span a range of starting conditions and investment intensities. The sample includes Copenhagen and Amsterdam at the high end of established cycling culture, Paris and Seville as cases of recent rapid investment, London and Berlin as large cities with mixed approaches, and Rome and Madrid as cases of modest investment. The remaining four cities are Helsinki, Bordeaux, Lyon, and Munich, each representing distinct geographic and demographic profiles.

2. Data and Methods

Modal share data was sourced from each city's most recent household travel survey, with figures reported as the percentage of all trips taken by bicycle. Where surveys were not conducted annually, linear interpolation was applied between the nearest available years. Capital expenditure data was extracted from municipal budget documents, with cycling-specific allocations identified through line-item review and cross-checked against published infrastructure delivery reports.

Two principal metrics were used. The investment intensity metric expresses cycling capital expenditure as a proportion of total transport capital expenditure over the ten-year period. The infrastructure density metric captures the kilometres of protected cycle lane per 100,000 residents, measured at the end of the study period. The modal share change is calculated as the percentage point difference between 2014 and 2024 figures.

Cities were classified into three investment categories: high (above 8 percent of transport capital expenditure), medium (between 3 and 8 percent), and low (below 3 percent). Statistical analysis was conducted using ordinary least squares regression, with controls for population growth, baseline cycling rate, and topographic gradient. Confidence intervals are reported at the 95 percent level throughout.

3. Findings

The headline finding is a strong positive relationship between investment intensity and modal share growth. Cities in the high-investment category, namely Copenhagen, Paris, Seville, and Bordeaux, exhibited an average modal share increase of 4.2 percentage points (95% CI: 3.1 to 5.3). Medium-investment cities, including Amsterdam, Helsinki, Lyon, and Munich, averaged 2.1 points (95% CI: 1.4 to 2.8). Low-investment cities, namely London, Berlin, Rome, and Madrid, averaged 0.8 points (95% CI: 0.2 to 1.4).

The relationship is non-linear with respect to infrastructure density. Cities below 50 kilometres of protected cycle lane per 100,000 residents exhibited modest growth regardless of investment intensity, while cities above this threshold experienced compounding gains. We interpret this as evidence of a network effect: a sparse network does not provide enough route coverage to make cycling a viable choice for most journeys, regardless of how much is spent on individual segments.

Paris is the most striking case in the sample. Between 2014 and 2024, Paris increased protected cycle lane kilometres from 18 to 84 per 100,000 residents and saw cycling modal share rise from 3 percent to 11.2 percent. Seville follows a similar trajectory at smaller scale: an investment programme concentrated in the years 2015 to 2019 produced a modal share increase from 5.6 percent to 9.1 percent. By contrast, London's modal share moved from 2.1 percent to 3.2 percent over the same period, despite high absolute spending in nominal terms, because that spending represented less than 2 percent of London's much larger total transport budget.

Topographic gradient was a statistically significant control variable. Cities with average gradients above 4 percent exhibited modal share gains 1.3 percentage points lower than otherwise comparable cities, holding investment intensity constant. This effect persists after the introduction of electric-assist bicycles, although the gap has narrowed in the most recent years of the study period.

4. Discussion

The findings should be interpreted with caution given the small sample size and the difficulty of isolating causal effects in observational data. Nonetheless, the magnitude of the differences between investment categories, and the consistency of the pattern across cities with different starting conditions, supports the interpretation that sustained capital investment is a meaningful driver of modal share change.

The non-linear network effect deserves particular attention from policymakers. Incremental investment that fails to cross the density threshold produces disappointing returns and may undermine political support for further investment, creating a self-reinforcing low-equilibrium trap. The implication is that cities seeking to grow cycling modal share should aim for concentrated investment over a defined period rather than dispersed incremental spending. Paris and Seville both adopted this concentrated approach and both achieved the strongest growth in the sample.

The London case is instructive. Despite ranking among the highest-spending cities in absolute terms, the proportional commitment was insufficient to drive meaningful modal share growth. This suggests that absolute spending is a misleading measure when comparing cities of different sizes and that proportional commitment, expressed as a share of overall transport budget, is the more reliable indicator. Future studies might investigate whether this pattern holds in non-European contexts.

5. Limitations

Several limitations should be acknowledged. First, household travel surveys differ across cities in methodology, sampling frame, and definitions; while we have applied harmonisation procedures, residual measurement error remains. Second, capital expenditure data does not capture operating expenditure on maintenance, programme administration, and behavioural change campaigns, which may be material in some cases. Third, the ten-year window may be too short to capture the full effect of investments made late in the period. Finally, the sample is restricted to European cities and the findings may not generalise to cities with different urban form, climate, or transport cultures.

6. Conclusion

Sustained capital investment in cycling infrastructure is associated with meaningful growth in cycling modal share, but the relationship is non-linear and depends on crossing a network density threshold of approximately 50 kilometres of protected cycle lane per 100,000 residents. Concentrated investment over a defined period appears more effective than dispersed incremental spending. Proportional commitment to cycling, measured as a share of total transport capital expenditure, is a more reliable predictor of outcomes than absolute spending. These findings should inform the design of transport investment programmes in cities seeking to expand cycling as a credible mode of urban travel.